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Course: BIOS331 – Microbial Genomics | [redacted]
Assignment: Final Project

Bacterial GWAS on Strains in Stool Samples of Colorectal Cancer Patients

Literature Review

The gut microbiome is linked to Colorectal Cancer (CRC) development, which has been supported through many studies and a substantial amount of evidence. An unbalanced gut microbiota, commonly termed dysbiosis, could result in inflammation, intestinal barrier failure, and oncogene upregulation, which all favor the development of CRC (Zhao et al., 2022). Several individual species of opportunistic bacteria in the gut microbiota have been constantly reported to be associated with CRC. For instance, *Fusobacterium nucleatum* has been widely studied as a CRC-promoting microbe in both human and animal models (Ternes et al., 2020; Zhao et al., 2022). Both *Prevotella intermedia* and *Parvimonas micra* have been linked to CRC at the species level. *P. intermedia* has been associated with higher CRC risk in African-American cohorts and detected in multiple international CRC metagenomic studies, suggesting a potential role in disease progression via immune modulation or genotoxic effects. *P. micra* has been found in abundance in CRC patient stools, highlighting its possible mechanistic involvement in CRC (Ternes et al., 2020).

Although the relationships between microbes and CRC at the species level are well established, they are merely associative and are unable to pinpoint the underlying causes. A major limitation is intra-species heterogeneity: not all strains of microbes are equally harmful. For example, Zepeda-Rivera et al. (2024) demonstrated that only a specific clade is consistently enriched in tumors, despite the fact that *Fuscobacterium nucleatum* has long been associated with CRC. The fact that other lineages of *F. nucleatum* are not enriched in tumors underscores the importance of strain-level resolution. This illustration shows that tumour tropism (when tumor cells spread and target specific organs of the body) is determined by genetic variation within species. A gene-level approach is motivated by the fact that although *P. micra* and *P. intermedia* are both enriched in CRC (Zhao et al., 2022; Lo et al., 2022), the contribution of particular strains or clades is still unknown.

To take one step ahead and move beyond species-level association to gene-level approaches, bacterial genome-wide association studies (bGWAS) is one of the methodological frameworks adopted. bGWAS are families of methods to statistically associate phenotypes with genotypes, such as point mutations and other variants across the genome (Saber & Shapiro, 2020). Over the past decade, microbial GWAS using SNPs and k-mers has successfully identified genetic determinants across diverse contexts — including antibiotic resistance in *Mycobacterium tuberculosis*, virulence traits in *Streptococcus pneumoniae* and *Staphylococcus aureus*, and host preference in *Campylobacter jejuni* (Saber & Shapiro, 2020).

While bacterial GWAS is a powerful tool, its ability to function is contingent on mutation load, recombination rates, and sample size. Studies pose the risk of false positives or insufficient power if these parameters are not taken into consideration. BacGWASim, a simulation framework that is different from bGWAS and is used for evaluating method performance, was created by Saber and Shapiro (2020). Their research showed that while clonal populations weaken power, species with greater diversity and recombination generate more trustworthy associations. In addition, the study showed that the choice of method affects the results, with linear mixed models showing better results in the presence of robust population structure. These observations inform our design: we can optimize statistical robustness and interpretability by matching techniques to the evolutionary dynamics of our strains.

The steady enrichment of specific bacterial species in CRC is well-documented but insufficient. It obscures important intra-species heterogeneity, which is the idea that pathogenicity is caused by individual strains rather than species as a whole. This was demonstrated in *F. nucleatum.* We apply this paradigm to other important key contributors, such as *P. intermedia* and *P. micra*. Our project moves beyond species-level association to pinpoint the precise genetic determinants of CRC pathogenicity using bGWAS and pangenomics.

Aims

Many gut species, such as *Parvimonas micra, Bacteroides fragilis, Prevotella intermedia,* and possibly *Gemella morbillorum* are reproducibly enriched in CRC patients, but the specific genetic factors underlying these associations remain unknown. Our first aim is to identify the genetic determinants within these strain-specific species, mobile elements, or structural variants that may drive CRC risk using *Fusobacterium nucleatum* as a positive control.

Second, we aim to test whether incorporating gene-level information improves the prediction of CRC compared with models based solely on species abundances, and to assess the trade-offs in accuracy, interpretability, and cost (Piccino et al., 2025; Orellana et al., 2023).

Finally, we will map significant loci to biological pathways, such as adhesion capsule remodelling, iron and heme uptake, nitrosation, protease activity, and metabolism. This will allow us to evaluate whether multiple CRC-associated species converge on shared functional mechanisms. By doing so, we will provide candidate pathways and genes for experimental validation and clarify the potential diagnostic lift of gene-level models (Zepeda-Rivera et al., 2024).

Methods

To achieve these aims, we will combine public CRC stool and tissue metagenomes with a prospective sample set, ensuring sufficient statistical power and reproducibility across independent populations. In addition, we will draw an open-source stool metagenome such as the Washington DC case-control study of 52 CRC patients and 52 matched controls, which has been widely used for benchmarking reproducibility in microbiome-cancer associations (Piccino et al., 2025; Gehrig et al., 2022; Vogtmann et al., 2016).

For each target species, we will construct a "pangenome" to catalogue core and accessory genes. This provides the foundation for linking genetic variation to disease associations (Zepeda-Rivera et al., 2024; Orellana et al., 2023). For *Fusobacterium nucleatum*, we will apply a phylogenetic filter to select strains relevant to the CRC phenotype before performing gene-level association tests (Zepeda-Rivera et al., 2024).

We will then conduct bacterial genome-wide association studies (bGWAS) to test whether gene presence or absence or sequence variants are correlated with CRC status. Acknowledging known limitations like population structure, we will employ a nuanced bGWAS approach. Linear mixed models will be applied to control for shared ancestry among strains and reduce false positives (Saber & Shapiro, 2020; Coll et al., 2022). The primary bGWAS analysis will be performed on *Parvimonas micra, Prevotella intermedia*, and *Bacteroides fragilis*, given their reproducible enrichment in CRC. *Fusobacterium nucleatum* will serve as a positive control to benchmark our methods. Analysis of *Gemella morbillorum* will be contingent on a sufficient number of high-quality strain carriers being identified in the cohorts. Clinical covariates, including age, sex, BMI, and medication use, will be included in models to disentangle microbial gene effects from host factors (Piccino et al., 2025).

Associated genes will subsequently be mapped to functional pathways using KEGG and MetaCyc modules, enabling the detection of cross-species convergence in mechanism. Lastly, we will benchmark predictive models to compare species-level versus gene-level features and quantify the diagnostic lift provided by gene-level resolution by AUROC/PR gains and better calibration (Piccino et al., 2025; Orellana et al., 2023).

Expected Outcomes

1. Identification of priority strains, if present: Extending the *Fusobacterium* paradigm (Zepeda-Rivera 2024) to the target genera
2. Catalogs of CRC-associated loci: Unitigs/genes/alleles with pan-genome context for *P. micra, B. fragilis, P. intermedia*, and, if enough carriers identified, *G. morbillorum*. We can further stratify by specific priority clades (Contingent on Outcome 1).
3. Better mechanistic understanding: We hope to delineate convergent metabolic axes across genera in the tumor niche (e.g., iron/heme uptake, nitrosative metabolism, proteases, etc).
4. Strain-aware diagnostic improvement: Current models predicting CRC from microbiome data use abundance metrics as inputs. We will test if adding compact gene-level features (presence/absence or allelic variants) improves prediction in external validation cohorts, informing next-generation stool tests and strain-targeted interventions.
5. Communication & reproducibility: We aim to publish the findings through manuscripts in high-impact peer-reviewed journals. Reproducible pipelines, workflows, and code with clear documentation will be released so the work can be repeated or extended by others. Work on this study's cohort and other data will further serve as a validation of previous *Fusobacterium* findings.

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